POLYPULSE — System Architecture

AI-powered longitudinal monitoring of endometrial polyps using wearable ultrasound and 3D analysis

From Scan to Insight

Continuous monitoring | Early risk detection
Personalized clinical support

1 USER INTERFACE & DATA INPUT

Capture, view and interact with ultrasound data

Wearable Ultrasound Device

Patient performs periodic scans at home / clinic



- Ultrasound images & video
- Device metadata
- Scan timestamp

Patient Web / Mobile App

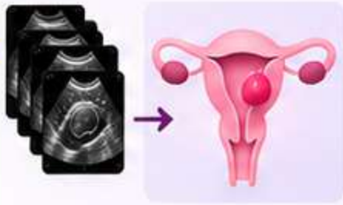
View scans, track progress, receive insights



- Scan upload
- 3D model view
- Progress dashboard
- Educational insights

AI Processing & 3D Reconstruction

Converts 2D ultrasound into 3D uterus and segments polyps



- Image enhancement
- Uterus reconstruction
- Polyp segmentation
- Size, volume & shape extraction

Patient Dashboard

Longitudinal tracking and easy visualization



- 3D interactive model
- Timeline comparison
- Size and volume trends
- Visual change indicators
- Personalized insights

2 BACKEND & CORE SERVICES

Secure data flow, AI analysis, storage and reporting

Data Ingestion Layer

Receives and validates ultrasound scans from app

- Secure upload (HTTPS)
- Metadata validation
- De-identification & encryption
- Store raw scans

AI & Image Processing Engine

Processes ultrasound data for analysis

- Image preprocessing
- Uterus and polyp segmentation
- 3D-reconstruction
- Measurement extraction (size, volume, morphology)

Database & Model Storage

Stores scans, 3D models and analysis results

- Encrypted storage (PostgreSQL)
- Versioned 3D models
- Patient longitudinal records
- Audit logs

Application Server & APIs

Handles business logic and data access

- REST APIs (FastAPI)
- User authentication and role management
- Data processing pipeline
- Serve data to web and mobile app

Doctor Dashboard & Alerts

Clinician views detailed analysis and receives automated alerts

- Patient-wise longitudinal reports
- AI-generated measurements
- Risk alert for significant changes
- Compare multiple timepoints
- Download reports for consultation

3 FUTURE INTEGRATIONS & EXPANSION

Planned capabilities to enhance research and clinical utility

EHR Integration

- Sync with hospital systems
- Import past records
- Streamline clinical workflow

Advanced AI Analytics

- Growth prediction models
- Risk stratification
- Personalized follow-up recommendations

Multi-Center Data & Research

- Anonymized data aggregation
- Population-level insights
- Support for clinical research

Teleconsultation Module

- Remote expert consultation
- Share 3D models and reports
- Integrated communication

Regulatory & Deployment

- Clinical validation
- Compliance and security
- Scalable cloud deployment